

Your grade: **100%**

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1. Which of the following are well-defined sets?

1 / 1 point

☒ All prime numbers less than 100

✔ Nice work

The set of prime numbers less than 100 is finite and well defined

☐ Good books in the library

☒ A group of students enrolled in discrete mathematics

✔ Nice work

This is a well-defined set

☐ None of these answers

2. Which one of the following sets are equal?

1 / 1 point

☒ $A = \{1, 2, 3\}$ and $B = \{3, 1, 2\}$

☐ $A = \{1, 2\}$ and $B = \{1, 2, 3\}$

☐ $A = \{1, 2, 3\}$ and $B = \{1, 2, 4\}$

☐ $A = \{1, 2\}$ and $B = \{1\}$

✔ Nice work

The order of elements in a set is not important

3. Let $S = \{\emptyset, \{a, b, c\}\}$. What is the cardinality of S?

1 / 1 point

☐ 4

☐ 3

☐ 0

☒ 2

✔ Nice work

$\emptyset$ and $\{a, b, c\}$ are the only elements of S , hence $|S| = 2$

4. Which of the following is a subset of $\{b, c, d\}$?

1 / 1 point

☐ $\{1, 2, 3\}$

☐ $\{a\}$

☒ $\emptyset$

☐ $\{a, b, c\}$

✔ Nice work

$\emptyset$ is a subset of any set

5. Which of the following statements are true?

1 / 1 point

☒ $\emptyset \subseteq \{\emptyset\}$

✔ Nice work

$\emptyset$ is a subset of any set

☒ $\emptyset \in \{\emptyset\}$

✔ Nice work

$\{\emptyset\}$ contains $\emptyset$, hence, $\emptyset \in \{\emptyset\}$

☐ $\emptyset \in \emptyset$

☐ $\{\emptyset\} \in \{\emptyset\}$